

EmoMoconvq: Emotion-Conditioned Human Motion Generation

Midterm Progress Report

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Abstract—Physics-based motion generation methods such as MoConVQ [7] produce high-quality motions from text but generate stylistically neutral results regardless of emotional context. This project, EmoMoconvq, extends MoConVQ’s text-to-motion transformer (T2M-MoConGPT) by injecting a discrete emotion label as an additional conditioning signal, learned through parameter-efficient fine-tuning. As a method-extension project, we build directly on the pretrained MoConVQ framework and add a trainable emotion module without modifying the original code. This report summarizes concrete progress: (i) the T2M-MoConGPT baseline has been reproduced and verified; (ii) an emotion-labelled text corpus has been curated from HumanML3D [2] by keyword filtering; (iii) the emotion embedding module has been implemented and verified to be a clean parameter superset of the pretrained checkpoint; and (iv) a from-scratch negative log-likelihood fine-tuning loop has been implemented and validated in a controlled experiment, where the loss decreases from 6.95 to $\sim 10^{-4}$ and the emotion module receives non-zero gradients throughout. We also report honest preliminary findings – a smaller-than-expected labelled set and a representation gap in the motion-tokenization pipeline – and how they shape the remaining plan.

I. PROBLEM STATEMENT AND MOTIVATION

Human motion is inherently expressive: the same action performed in different emotional states (e.g., walking while happy vs. sad) exhibits distinct kinematic patterns such as stride length, torso posture, and arm-swing amplitude. This expressiveness is critical for digital humans, game characters, and social robots, yet current physics-based generation methods ignore it entirely.

MoConVQ [7] shows that discrete residual-VQ representations combined with a GPT-style transformer can generate diverse, physically accurate motions from natural language. However, its T2M-MoConGPT produces emotion-neutral outputs: the same text prompt always yields the same stylistic result. The robot-learning problem we address is therefore *how to condition a pretrained generative motion model on a high-level affective signal while preserving the physical plausibility guaranteed by its simulator*. We address it with EmoMoconvq, which extends T2M-MoConGPT with discrete emotion conditioning. The project scope is unchanged from the proposal; the refinements below (a feature-dimension correction and a newly identified data-pipeline gap) are implementation-level findings rather than a change of direction.

II. RELATED WORK

Physics-based motion generation. MoConVQ [7] encodes motions into residual VQ codes and decodes them through a physics simulator via model-based RL, trained on 23 hours of

motion data. Its frozen encoder and codebook make it well suited for lightweight downstream fine-tuning, which is the basis of our work.

Text-to-motion generation. T2M-GPT [8] and MDM [6] generate kinematic motions from text using discrete transformers and diffusion models respectively, both trained on HumanML3D [2]. Neither is physics-based nor models emotional style.

Style and emotion in motion. ACTOR [4] conditions a Transformer VAE on action labels, showing that categorical signals can guide motion style. The CMU MoCap database [1] contains emotionally performed locomotion. Our work is, to our knowledge, the first to combine emotion conditioning with a physics-based generation pipeline. For parameter-efficient adaptation we draw on the adapter/LoRA family [3].

III. TECHNICAL APPROACH

A. System Overview

We build on the pretrained MoConVQ framework [7], which has three trained components: (1) a 1D-convolutional **motion encoder** $\mathcal{E}$ mapping a motion clip to latent vectors; (2) a **residual VQ codebook** $\mathcal{B}$ ($N = 8$ quantization layers, 512 codes of dimension 768) producing discrete index sequences; and (3) a **physics-based decoder** $\mathcal{D}$ (deconvolution + mixture-of-experts policy + simulator). For text-to-motion, MoConVQ uses a dual transformer **MoConGPT**: a *temporal transformer* ($N_T = 12$ cross-attention layers) that aggregates information across time, and a *depth transformer* ($N_D = 5$ layers) that predicts the RVQ indices layer by layer within each time step. We freeze all of these and introduce a trainable emotion module plus a small number of top temporal-transformer layers.

B. Emotion Embedding Module

Let $\mathcal{C} = \{\text{neutral}, \text{happy}, \text{sad}, \text{angry}, \text{fearful}\}$, $|\mathcal{C}| = 5$. Each emotion is mapped to a trainable vector

$$E_e : \mathcal{C} \rightarrow \mathbb{R}^{d_e}, \quad e_c = E_e(c), \quad (1)$$

with $d_e = 512$. It is fused with the text feature through a two-layer MLP and element-wise addition:

$$f_{\text{cond}} = f_{T5} + \text{MLP}_{\text{proj}}(e_c). \quad (2)$$

Dimension correction. The proposal stated d_e matches the temporal transformer hidden size; on inspection, the feature consumed by the cross-attention layers is the 1024-dimensional T5-large [5] text feature f_{T5} . Hence MLP_{proj} :

$\mathbb{R}^{512} \rightarrow \mathbb{R}^{1024}$, and fusion occurs in the text-feature space, immediately after the text encoder and before the temporal transformer’s cross-attention blocks. The fused f_{cond} replaces f_{T5} in those layers, exactly the original injection mechanism.

Graceful-degradation initialization. The last layer of MLP_{proj} is zero-initialized, so at the start of fine-tuning $\text{MLP}_{\text{proj}}(e_c) = \mathbf{0}$ and $f_{\text{cond}} = f_{T5}$: the model is numerically identical to the pretrained baseline and can only improve from there. This design is verified empirically in Section IV. Figure 1 summarizes the architecture.

C. Fine-Tuning Strategy

We freeze the encoder, codebook, physics decoder, depth transformer, the text encoder, and the bottom $(N_T - K)$ temporal-transformer layers. Only the emotion table E_e , the projection MLP_{proj} , and the top $K = 4$ temporal layers are trained, with the same negative log-likelihood (NLL) objective as MoConGPT:

$$\mathcal{L}_{\text{emo}} = \mathbb{E}_{(S,c)} \left[\sum_k \sum_d -\log p(I_d^k \mid f_{\text{cond}}(c), I_{<d}^k, I^{<k}) \right]. \quad (3)$$

The official MoConVQ release does not include MoConGPT training code, so this loop is implemented from scratch following the teacher-forcing structure of the model. The emotion module is realized as a non-invasive subclass (`EmoText2Motion_Transformer`); the original MoConVQ source is left unmodified, which keeps the baseline runnable for ablation and makes our contribution explicit. Parameter counts are reported in Table I.

D. Data Preparation

Following the proposal, we filter the HumanML3D [2] training split for captions containing emotion-related adverbs/adjectives (e.g., “happily”, “sadly”, “angrily”, “fearfully”), grouped into four non-neutral classes, and assign labels by keyword matching. Captions matching two classes are treated as conflicts and discarded.

IV. PRELIMINARY EXPERIMENTS AND RESULTS

A. Baseline Reproduction

The pretrained T2M-MoConGPT pipeline was installed and reproduced. From the text prompt “A person walking while talking on the phone” the system generated a physically simulated motion of 840 frames at 120 fps (7 s). Figure 2 shows representative frames of the simulated character, confirming the baseline produces plausible, physically valid motion and that our environment is correct.

B. Emotion Dataset Curation

The keyword filter was applied to the 23,384 sequences (69,896 captions) of the HumanML3D training split, yielding **463** emotion-labelled caption–motion pairs (69,431 neutral, 2 conflicts discarded). The class distribution is shown in Figure 3. Three findings emerge and directly inform the plan:

TABLE I
EMOTION MODULE VERIFICATION AND PARAMETER BUDGET.

Check / Quantity	Result
Missing emotion tensors on checkpoint load	5
Missing non-emotion tensors / unexpected keys	0 / 0
$ \text{out}(\emptyset) - \text{out}(\text{happy}) $ at init (zero-init)	0.0
$ \text{out}(\text{happy}) - \text{out}(\text{sad}) $ after perturbation	1.40
Emotion module parameters	0.79M
Trainable parameters (emotion + top-4 layers)	47.3M
Total model parameters	194.4M
Trainable fraction	24.3%

- The yield (463) is *below* the proposal’s expected 800–1200: HumanML3D captions are predominantly literal action descriptions and rarely contain affective words.
- The classes are strongly imbalanced (*sad* has only 28 pairs), which triggers the proposal’s backup plan (class reduction, augmentation, and the CMU emotional-gait supplement).
- HumanML3D already ships left/right-mirrored copies; 229 of 463 pairs are mirrors. Thus the proposed mirroring augmentation is effectively built in, and the remaining lever is temporal speed augmentation ($0.8 \times / 1.2 \times$, $\approx 3 \times$ expansion).

C. Emotion Module Verification

The emotion module was validated by five structural checks (Table I). Loading the pretrained checkpoint into `EmoText2Motion_Transformer` with `strict=False` leaves exactly the 5 new emotion tensors uninitialized and produces *no* unexpected keys, confirming the module is a clean parameter superset of the baseline. With the zero-initialization, the temporal feature with and without an emotion label is bit-identical at step 0 (max difference 0.0), verifying graceful degradation; after perturbing the projection, *happy* and *sad* produce clearly different outputs (max difference 1.40), confirming the conditioning pathway is functional. The configuration is parameter-efficient: the emotion module is only 0.79M parameters, and the full trainable set is 47.3M of 194.4M (24.3%).

D. Initial Fine-Tuning

Because the HumanML3D motions are SMPL-based kinematic vectors while the MoConVQ encoder consumes physics-character observations, the labelled corpus above does not yet carry MoConVQ motion tokens (see Section V). To still validate the fine-tuning pipeline end-to-end, we ran a *controlled overfitting experiment*: the one MoConVQ-format motion shipped with the pretrained data (a CMU walking clip, 5227 frames) was encoded into RVQ tokens (8×1306), sliced into five 50-step clips, and each clip was paired with an emotion-labelled caption. The emotion-conditioned model was then fine-tuned with the NLL loss of Eq. 3 for 60 steps.

Figure 4 shows the NLL dropping from 6.95 to $\sim 10^{-4}$, while the emotion-module gradient norm is large early and

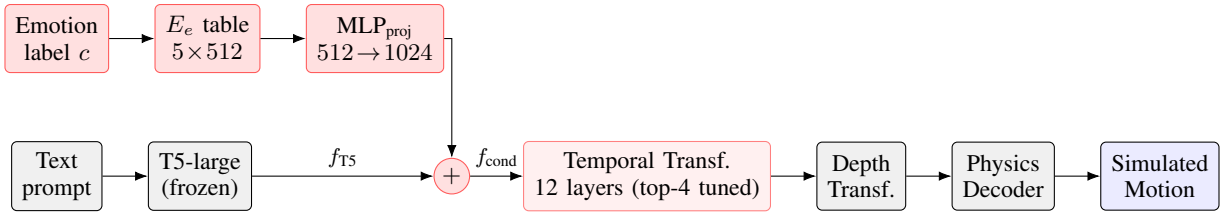


Fig. 1. EmoMoConvq architecture. Red blocks are trainable (the emotion module and the top-4 temporal-transformer layers); grey blocks are frozen pretrained MoConVQ components. The emotion signal is injected additively into the T5 text feature (Eq. 2).

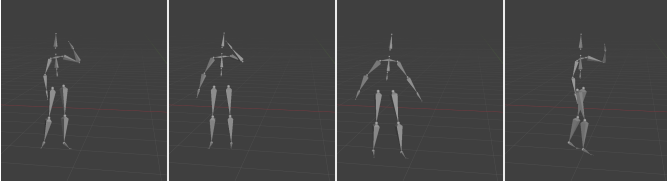


Fig. 2. Reproduced T2M-MoConGPT baseline: frames from a physically simulated motion generated from a text prompt. The output is stylistically neutral, motivating emotion conditioning.

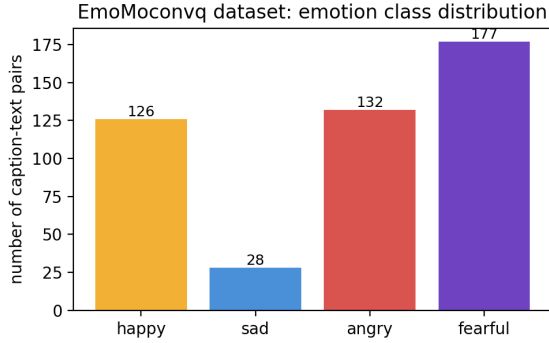


Fig. 3. Emotion class distribution of the 463 caption-motion pairs filtered from the HumanML3D training split.

decays as the model converges – the emotion module is genuinely trained (its embedding table moves by $L_2 = 0.076$). An *emotion-swap* diagnostic (Table II) re-evaluates each clip with a wrong emotion label: 2 of 5 clips (*neutral*, *happy*) show a clear loss increase, confirming the model does route information through the emotion signal; the other 3 do not, because the teacher-forced latents alone already suffice for those clips. This is an honest, expected outcome of a tiny controlled set: it validates that **the fine-tuning loop and the emotion module are correct and trainable**, while a genuine test of emotion controllability requires the full dataset.

V. REMAINING WORK AND UPDATED PLAN

The experiments above place the project clearly past the proposal stage: the baseline, the dataset pipeline, the emotion module, and the fine-tuning loop are all implemented and verified. The main remaining work is:

(1) Motion-tokenization pipeline. The key gap is bringing HumanML3D motions into MoConVQ’s token space. HumanML3D motions are SMPL-based 263-d kinematic vec-

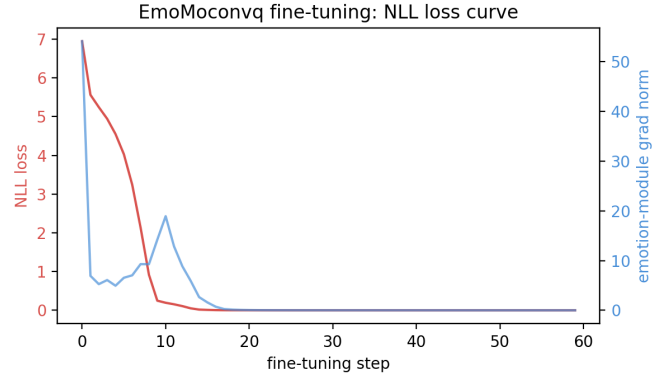


Fig. 4. EmoMoConvq fine-tuning. NLL loss (red) and emotion-module gradient norm (blue) over fine-tuning steps in the controlled overfitting experiment.

TABLE II
EMOTION-SWAP DIAGNOSTIC: NLL WITH THE CORRECT VS. A WRONG EMOTION LABEL AFTER FINE-TUNING (LOWER IS A BETTER FIT).

Clip	Correct label	Wrong label
neutral	0.0001	1.338
happy	0.0001	0.174
sad	0.0001	0.002
angry	0.0001	0.000
fearful	0.0001	0.000
mean	0.0001	0.303

tors, whereas the MoConVQ encoder consumes physics-character observations. We will build a retargeting path (HumanML3D/AMASS $\rightarrow$ BVH $\rightarrow$ MoConVQ character via the existing tracking utility), producing the (text, emotion, tokens) triples needed for real fine-tuning.

(2) Full-scale fine-tuning on the 463-pair set with temporal-speed augmentation, supplemented by CMU emotional-gait clips for under-represented classes (especially *sad*); fall back to 3 classes if overfitting occurs.

(3) Quantitative evaluation: FID, R-precision, an emotion-classifier accuracy metric, and a small user study, as specified in the proposal.

Two practical issues were also resolved: a GPU-architecture mismatch (the RTX 50-series is not natively supported by the available PyTorch build, which falls back to PTX JIT) and the absence of public MoConGPT training code (the loop was

TABLE III
UPDATED PROJECT TIMELINE. TASKS ABOVE THE LINE ARE COMPLETE.

Status	Milestone
Done	Baseline reproduced; environment verified.
Done	Emotion dataset curated; statistics analysed.
Done	Emotion module implemented and verified.
Done	NLL fine-tuning loop implemented and validated.
Next	Build HumanML3D→MoConVQ tokenization.
Next	Full-scale emotion fine-tuning + augmentation.
Next	Quantitative evaluation and user study.
Final	Report and demonstration video.

reimplemented). The updated timeline is given in Table III.

REPRODUCIBILITY AND ATTRIBUTION

This project extends the pretrained MoConVQ [7] framework. All EmoMoconvq code – the emotion module, the dataset filter, the tokenization script, and the fine-tuning loop – is original and contained in new files; the upstream MoConVQ source is unmodified. Pretrained weights and the HumanML3D [2] annotations are used under their respective terms.

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